

Optional Exercises I

*Instructor: Robert Rand***1 Recurrence Relations****Problem 1:** Solve the recurrence relation

$$H_n = 3H_{n-1} - 1$$

where $H_0 = 1$. Use the substitution method.**Problem 2:** Solve the recurrence relation

$$H_n = \frac{1}{2}H_{n-1} + 2$$

where $H_0 = 2$. Use the iteration method.**Problem 3:** Solve the linear recurrence relation

$$a_n = 3a_{n-1} + 4a_{n-2}$$

where $a_0 = a_1 = 5$.